

# Miquel Martí Florences

PHD RESEARCHER · CONTROL AND ENERGY SYSTEMS ENGINEER

Barcelona, Spain

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*Focused resume version for tailored applications in research, control, and energy systems engineering.*

## Summary

Control and energy systems engineer with research and laboratory experience in lithium-ion batteries, hybrid power platforms, and state estimation. Brings a combination of hands-on hardware work, model-based analysis, and scientific writing that fits both R&D and applied engineering roles.

## Experience

### CSIC-UPC, Institut de Robotica i Informatica Industrial (IRI)

Barcelona, Spain

PHD CANDIDATE & POSTDOCTORAL RESEARCHER

2023 - Present

- Conduct research on lithium-ion battery state estimation and control-oriented modeling, with a focus on regression-based estimation of internal variables.
- Develop and analyze estimation methods for state of charge and open-circuit voltage under challenging excitation and measurement conditions.
- Contribute to 3 journal publications and 2 submitted manuscripts in battery modeling, estimation, and related energy-system topics.

### CSIC-UPC, Institut de Robotica i Informatica Industrial (IRI)

Barcelona, Spain

RESEARCH ENGINEER

2021 - Present

- Maintain and upgrade a hybrid experimental station integrating fuel cells, supercapacitors, and lithium-ion batteries for control and energy research.
- Design control-related hardware, sensor interfaces, and TCP-based communication components to support real-time station operation and data acquisition.
- Contribute to research projects including DOVELAR and Hidrogena Renovable, supporting experimental development and laboratory integration.

### UPC Vilanova E3 Team

Vilanova i la Geltrú, Spain

HEAD OF ELECTRONICS AND POWERTRAIN

2016 - 2021

- Student team designing and building electric motorcycles for the MotoStudent competition.
- Designed and integrated the complete electrical system architecture of the electric motorcycle.
- Sized and modeled the battery pack from race energy requirements to support powertrain design decisions.
- Led a 6-7 person electronics and powertrain department, coordinating technical decisions, task execution, and subsystem integration.

### Applus+ IDIADA

Santa Oliva, Spain

E-MOTOR TEST ENGINEER INTERN

2019 - 2020

- Supported durability testing activities for electric motor components, contributing to automotive validation work.

### AKO Group

Vilanova i la Geltrú, Spain

FIRMWARE TEST ENGINEER INTERN

2019

- Automated firmware test processes in Python for gas detector products, improving testing repeatability and reducing manual effort.

## Education

### UPC-ETSEIB

Barcelona, Spain

PHD IN AUTOMATION, ROBOTICS AND VISION

2023 - 2026

- Research on regression-based estimation of internal variables in lithium-ion batteries, with emphasis on state-of-charge, model parameter and open-circuit-voltage estimation under challenging excitation conditions.

### UPC-ETSEIB

Barcelona, Spain

MASTER'S DEGREE IN INDUSTRIAL ENGINEERING

2020 - 2023

- Master's thesis: control architecture for a microgrid composed of a fuel cell, a battery, and a supercapacitor. Grade: 10/10.

### UPC-EPSEVG

Vilanova i la Geltrú, Spain

BACHELOR'S DEGREE IN INDUSTRIAL ELECTRONICS AND AUTOMATION ENGINEERING

2016 - 2020

- Bachelor's thesis: control analysis and EV application of a hybrid storage system based on lithium batteries and supercapacitors. Grade: 9/10.

## Skills

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|------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Battery &amp; Control</b> | Lithium-ion battery modeling, state estimation, observers, control systems, hybrid energy systems                |
| <b>Programming</b>           | Python, MATLAB/Simulink, C (small applications), LaTeX                                                           |
| <b>Lab &amp; Hardware</b>    | LabVIEW, cRIO (real-time and FPGA), sensor integration, Altium (small PCBs), oscilloscope, multimeter, soldering |
| <b>Languages</b>             | Catalan (native, C1), Spanish (native), English (C1), French (B1), German (A2)                                   |